REFLOWECONOMY · DESIGN PRECIS

ReflowEconomy design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-25	Amish Chadha	Kit 1.9.1 upgrade: concept option A of RFE-OPT-001 (choice awaiting Amish); STEM check fixes (0.88 kWh/kg, 120 mm feed slot); supply counted as currents (about 55 A against 40 A, RFE-DEC-001 D1); result +$0.17 per shift; detailed D3 model; PR numbers of RFE-PRS-001
0.7	2026-10-02	Amish Chadha	Approved follow-ups: cranked feed hood passes the reach check; tray crossing padded (2.02 m headroom); equipment $25,505, result +$0.57 per shift; RFE-DWG-001 Rev P4
0.6	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried in: chute reach check now, self-closing door and gates, first partner candidate
0.5	2026-09-30	Amish Chadha	Design for construction (RFE-DDR-003): equipment $25,300 against a $25,000 value-engineering target, result +$0.73 per shift, aisle 1.57 m, RFE-DWG-001 Rev P3, link to the build plan
0.4	2026-09-25	Amish Chadha	

			Recommendations accepted by Amish (DDR-002). 1.5 kW fume fan (maximum demand 8.75 kW, equipment $24,200, result +$1.61 per shift); intake quality rule; derated 5 kW press confirmed as reference; rented shed preferred; RFE-DWG-001 Rev P2
0.3	2026-09-25	Amish Chadha	Apply RFE-DDR-001 decisions (design choices no longer proposed); numbers replaced by RFE-CAL-001; layout from the parametric model and RFE-DWG-001; process recipes, economics model and passport schema v0.2 linked; safety updated (shredder enclosure, heater interlock, hoods)
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (reference micro-factory, material flow, first-order numbers, passport attachment points, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

ReflowEconomy design precis

ReflowEconomy is an open playbook for local micro-factories, built around one reference design: a plastics micro-factory on 59.4 m² (a rented shed, or two 40 ft container footprints) that takes in 100 kg of collected material per shift, sorts out metals, e-waste and residue, and turns HDPE and PP into sheets and beams while selling PET as clean flake. The calculation note RFE-CAL-001 shows, on paper, that 51.6 % of input mass is kept local as products and clean flake, 10 % is exported for industrial refining, 10 % is sold to a local mill and 28.4 % goes to licensed disposal, using 0.88 kWh per kilogram of output. Counted as currents, its morning loads draw about 55 A on paper from the 40 A single-phase supply it was designed for, so how the site gets its power is proposed to Amish (RFE-DEC-001 D1). The equipment costs $25,505 (indicative, excluding the building; $505 over the $25,000 value-engineering target). The site only breaks even at the assumed prices (about +$0.17 per shift before rent), so the product price, not the machinery, decides whether a site pays its way. A material passport opens at intake and travels with every lot that leaves.

This is option A, the reference micro-factory, of the concept options RFE-OPT-001; the alternative is a sort-and-bale hub (option B). B outscores A in that document, and the choice is proposed, awaiting Amish (RFE-DEC-001 D4). Every claim here was checked against known science in the STEM check RFE-STM-001. The model is detailed to D3 (`cad/src/detail_a.py`): every part from real stock with its fixings.

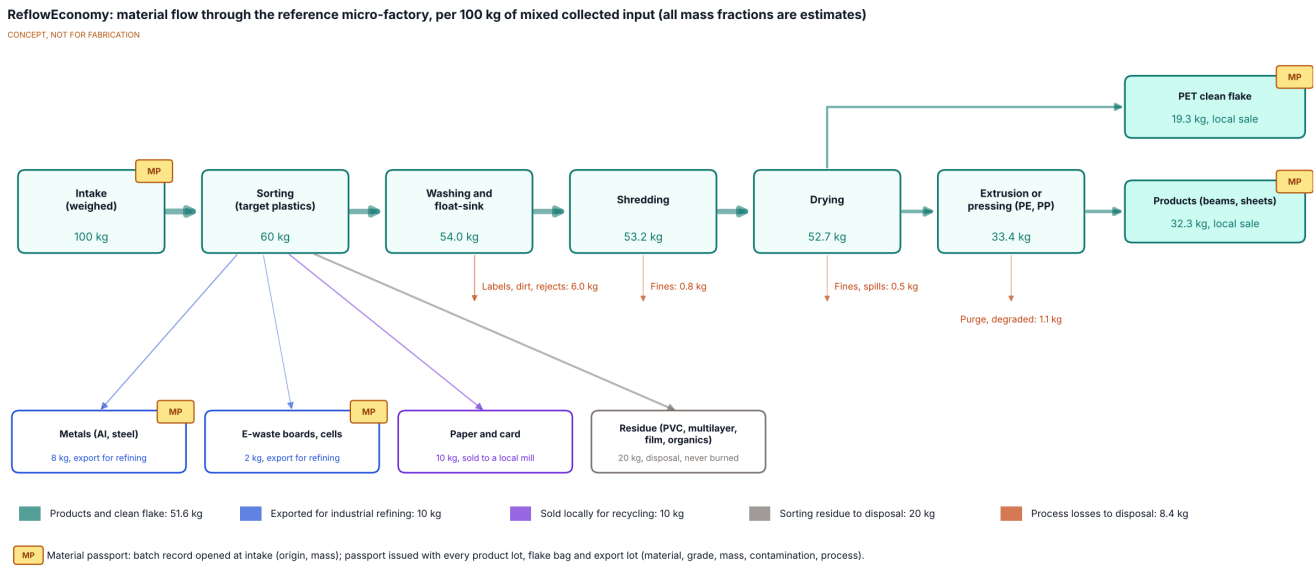


Figure 1. Material flow through the reference micro-factory per 100 kg of mixed collected input, from RFE-CAL-001. MP marks where a material passport record is opened (intake) and issued (every lot that leaves). All mass fractions are estimates.

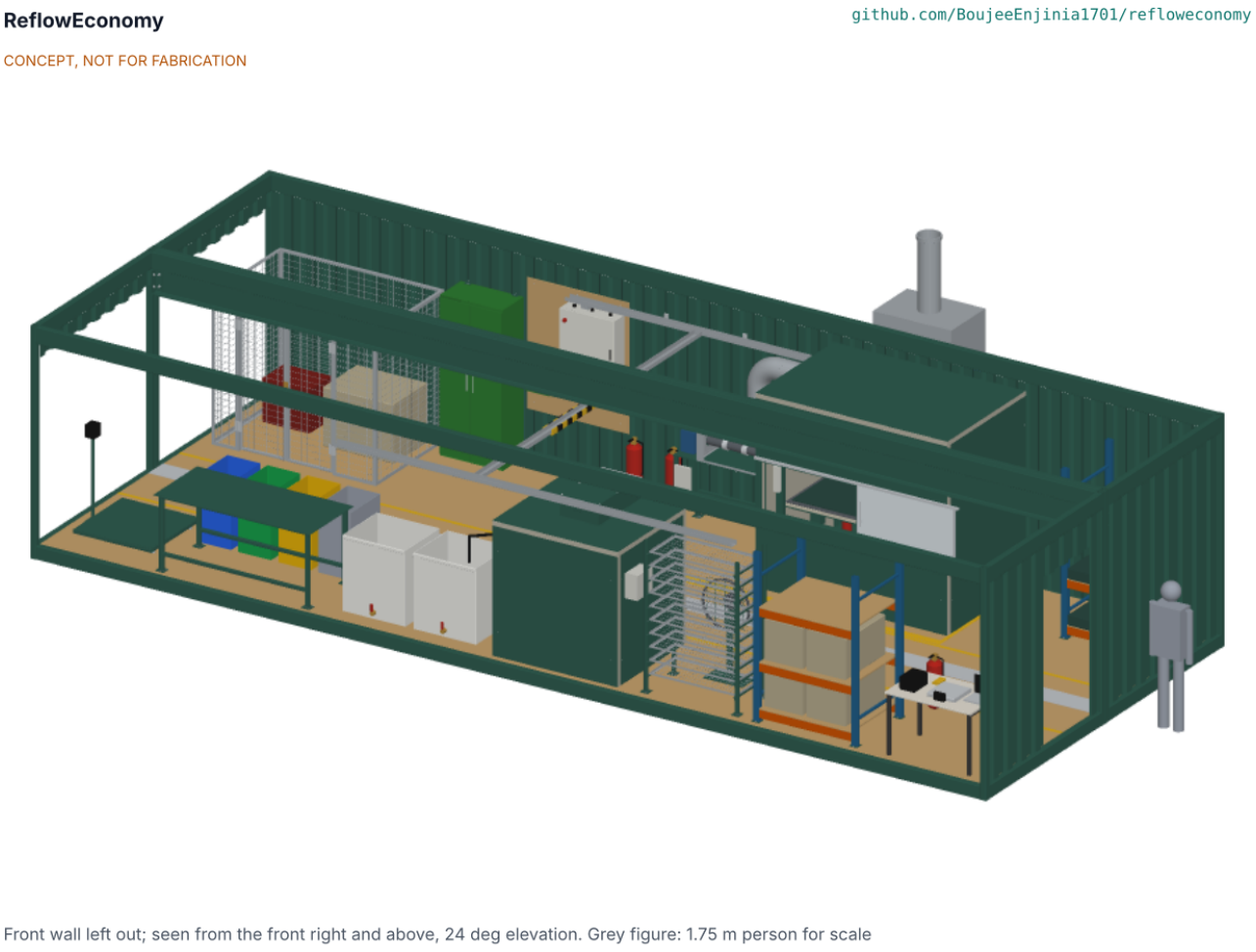


Figure 2. The reference micro-factory from the parametric model (`cad/src/model.py`) with the front wall left out and a 1.75 m person for scale. Material moves left to right from the intake door along a central aisle; the hot zone sits in the back row under two enclosing hoods.

What the playbook contains

Part	File	State at TRL 3
Material feasibility matrix	`docs/playbook/feasibility-matrix.md`	Draft, 7 material groups; hazard, export point and source columns still to add (PR-001)
Safety and environmental rules	`docs/playbook/safety.md`	Ten rules; expanded in the safety section below
Process recipes	`docs/playbook/recipes/`	PET flake, HDPE, PP and aluminium (add-on bay), each sourced (PR-002)
Economics model	`docs/playbook/economics_model.py` with `economics_inputs.csv`	Operator-editable, per kilogram and per shift, with break-even points (PR-003)
Reference micro-factory	This precis, `cad/src/model.py`, RFE-DWG-001, `bom/bom.csv`, RFE-CAL-001	Parametric layout, floor plan GA, priced equipment list and sizing
Material passport	`standards/material-passport-v0.2.schema.json`, `standards/examples/`	Schema v0.2 with example records; v0.1 kept

Machines live in their own repos: wastewise-scan identifies plastic resin at intake, WasteWise-ml classifies material class and grade from a phone photo, and a shredder, an aluminium micro-foundry and a cell tester are planned.

How it works

1. **Intake.** Pickers and collectors deliver bags at the intake door. Each delivery is weighed on the platform scale and gets a batch record: batch ID, origin (collection point), mass and date. This opens the material passport. A sample is sorted at once: a load with more than 10 % non-recoverable residue by mass is refused or charged a disposal fee (intake quality rule, PR-016). The picker is paid by weight and grade.
2. **Sorting.** At the sorting table, material is split by hand, helped by WasteWise-ml on a phone or a WasteWise Scan NIR reader for unclear plastics. Target plastics go to color-coded bins (PET, HDPE, PP). Aluminium and steel go to a metals bin, paper and card to bags, and circuit boards and cells to the export cage. PVC, multilayer packaging, film, polystyrene and organics go to residue. Lithium cells are pulled out first and stored in a fire-safe container.
3. **Washing.** Whole bottles and rigid parts are soaked, scrubbed and rinsed in the first tank, and labels are stripped.
4. **Shredding.** A guarded 2.2 kW shredder in a lined acoustic enclosure cuts clean parts into flake of about 5 to 10 mm, one polymer at a time.
5. **Float-sink and drying.** In the second tank, PP and HDPE flake floats and PET sinks, which removes cross-contamination. Flake dries on mesh trays with a fan. PET flake is bagged and sold as clean feedstock.
6. **Extrusion and pressing.** HDPE and PP flake dried on the previous shift becomes beams in the extruder (in the morning, while the shredder runs) and 1 x 1 m sheets in the sheet press (in the afternoon). A heater interlock keeps the press heaters off while the shredder or extruder runs, so the line works on a 40 A single-phase supply. Trim goes back to the shredder.
7. **Product store and passport.** Every product lot, flake bag and export lot gets a passport at the desk (schema v0.2: material, grade, mass, contamination, process, parent batch IDs and a recycled content statement) and a printed label, then goes to the racking or the export cage.

The steps are written out per material in the process recipes (`docs/playbook/recipes/`). Numbers match the exploded view (Figure 3), the floor plan RFE-DWG-001 and `bom/bom.csv` .

#	Zone or item	Choice	Notes
1	Intake scale and sorting table	300 kg platform scale, 2.0 x 0.9 m steel table, four color-coded bins (PET, HDPE, PP, metals)	Opens the batch record
2	Washing tanks and sediment trap	Two 400 to 500 L polyethylene tanks (pre-wash, float-sink), 0.75 kW pump, settling trap	Wash water reused; sludge to licensed disposal
3	Shredder with acoustic enclosure	Open-design shredder (Precious Plastic Shredder Pro class), 2.2 kW gearmotor on a single-phase input drive, interlocked feed chute with a cranked feed hood, lined enclosure with a self-closing door	15 kg/h assumed; about 15 dB insertion loss assumed
4	Drying rack and fan	Ten mesh trays (7.2 m²), 0.4 kW fan	7.3 kg/m² at the reference input
5	Extruder	Open-design extruder, 1.5 kW motor and 2.0 kW heaters, with barrel enclosure (hood B)	HDPE and PP only; 5 kg/h assumed
6	Sheet press and cooling press	1 x 1 m press with 5 kW single-phase heaters and a cooling press, in booth hood A	A derated variant of the published 15 kW, 400 V sheetpress
7	Fume extraction hoods and filter	Booth A (1.2 x 0.6 m sash) and hood B (0.8 x 0.4 m), 250 mm duct, 1.5 kW fan outside, G4, F7 and impregnated carbon filters, stack above the roof	0.57 m³/s at about 815 Pa; fan 66 % loaded with dirty filters (RFE-CAL-001)
8	Product store racking	Two bays: products in the back row, PET flake bags in the front row	Bay 1 is 1.0 m from the hot zone
9	Passport and quality desk	Desk, label printer, 30 kg bench scale, moisture meter, phone or laptop	Issues passports
10	Safety and PPE station	PPE cabinet, eyewash, first aid, three fire extinguishers	Entrance, hot zone and product end
11	Electrical board	230 V 40 A single-phase board, 30 mA RCDs, emergency stop circuit, heater interlock, energy meter	Installed by a licensed electrician
12	Export and residue cage	Lockable mesh cage by the export door, fire-safe cell container, bales and bags	Holds lots awaiting collection
13	Building shell	Preferred: a rented shed of about 60 m². Otherwise two 40 ft containers side by side (12.19 x 4.88 m) with the adjoining walls removed to a structural engineer's design	Site-dependent; excluded from the equipment total

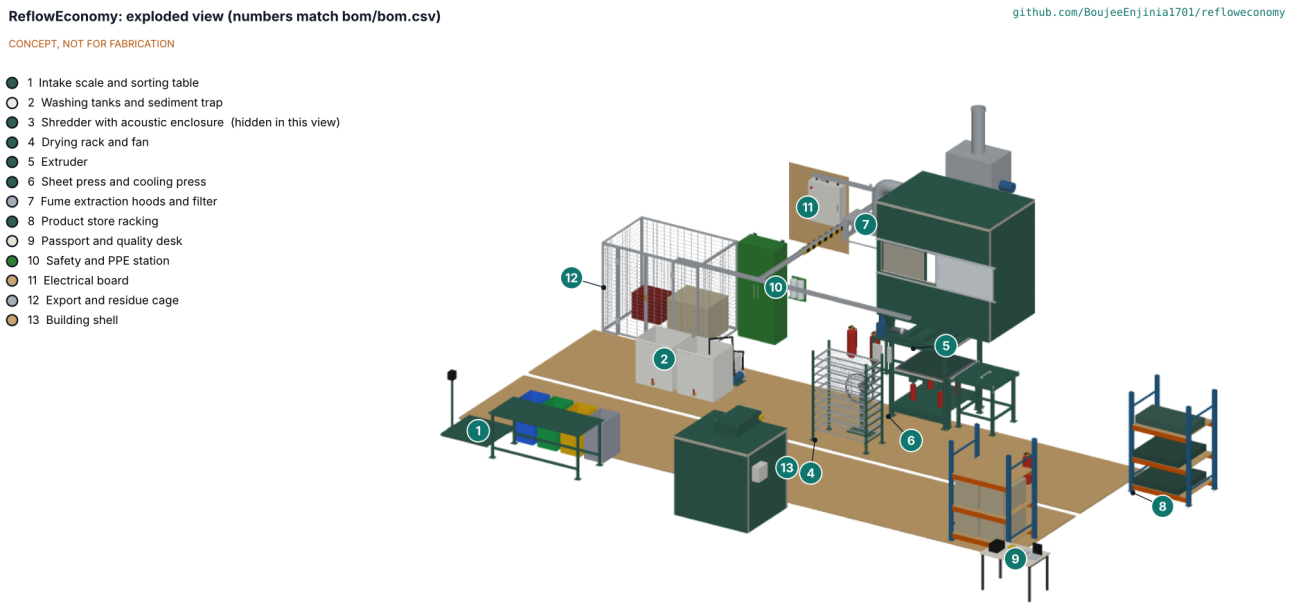


Figure 3. Exploded view of the reference layout with numbered zones matching `bom/bom.csv`*. The walls are omitted so nothing is hidden.*

Layout

The floor plan general arrangement is RFE-DWG-001 Rev P4 (`cad/drawings/RFE-DWG-001.pdf`), generated from `cad/src/model.py`. A central aisle runs along the middle of the floor (the seam in the container option), 1.20 m painted and 1.57 m clear between equipment, with 2.02 m headroom under the padded and taped cable tray crossing. How each part is made and fitted is in the prototype build plan RFE-BLD-001 (`docs/05-build-plan.md`); the changes that made the layout buildable are in RFE-DDR-003. The front row holds intake, washing, the enclosed shredder, drying, the flake racking and the passport desk; the back row holds the export cage by the export door, the safety station, the electrical board, the hot zone and the product racking. Doors are at the intake end (intake and export) and a personnel exit is at the far end.

Numbers from RFE-CAL-001

All values are estimates on paper. The calculation note gives the assumptions and the script `docs/04-calcs/sizing.py` prints every figure.

Table 1. Key results per shift of 100 kg input.

Quantity	Value	Requirement
HDPE and PP products; PET clean flake	32.3 kg; 19.3 kg (51.6 % kept local)	PR-007 met
Metals and e-waste exported for refining	10.0 kg	PR-004 not met (per-stream rules to write)
Paper and card sold to a local mill	10.0 kg	
To licensed disposal	28.4 kg (20.0 kg sorting residue, 8.4 kg process losses); 19.5 kg for a load at the 10 % intake threshold	**PR-008 not met** for the reference input; PR-016 met
Energy	45.6 kWh, 0.88 kWh/kg of output (motor and drive losses counted)	PR-009 met
Supply current, morning block	About 55 A running (72 A at full load) against a 230 V, 40 A single-phase supply; afternoon 36 A	**PR-017 not met**; RFE-DEC-001 D1
Shredder time	3.79 h of 4 h at an assumed 15 kg/h	PR-006 at risk
Sheet press	2.06 sheets of capacity for 2 in the afternoon	PR-006 at risk
Fume extraction	0.57 m³/s, 815 Pa, 1.00 kW input on a 1.5 kW fan	PR-011 hoods sized; exposure not verifiable at TRL 3
Shredder noise at the operator	94 dB(A) bare, 79 dB(A) enclosed (LEX,8h, assumed sound power)	PR-012 at risk
Equipment, items 1 to 12	$25,505 indicative	PR-010: $505 over the $25,000 value-engineering target
Result per shift before rent	+$0.17; break-even product price $2.49/kg; break-even input 100 kg	PR-003 met (model delivered)

The TRL 2 figures in v0.2 of this precis (30 kg products, 20 kg flake, about 45 kWh, about 13 kW peak, $22,850, margin about $18) are replaced by Table 1. The v0.3 figures changed by RFE-DDR-002 were 8.35 kW (36 A), $24,050 and +$1.73 per shift.

Economics

The operator-editable model (`docs/playbook/economics_model.py` , inputs in `docs/playbook/economics_inputs.csv`) gives an operating margin of about $21 per shift and a result of about +$0.17 after recovering the equipment over five years, before rent. Each $0.50/kg on the product price moves the result by about $16 per shift, so product design and local buyers matter more than machine cost. Prices and wages vary widely by country and must come from the co-design partner.

Material passport

The passport is a JSON record, one per lot, validated against `standards/material-passport-v0.2.schema.json` . It opens at intake as a batch record and is issued with every lot that leaves (Figure 1). Schema v0.2 adds `schema_version` , `product_form` , `parent_batch_ids` and a recycled content statement aligned with ISO 14021

(decided by Amish, 2026-09-25, RFE-DDR-001 item 4). Schema v0.1 (`standards/material-passport.schema.json`) is kept unchanged. Example records are in `standards/examples/` ; this one uses invented values:


```
{"schema_version": "0.2", "batch_id": "RFE-DEMO-S-0001", "product_form": "sheet",
 "parent_batch_ids": ["RFE-DEMO-I-0001", "RFE-DEMO-I-0002"], "material": "HDPE",
 "mass_kg": 11.2, "origin": {"country": "XX", "collection_point": "Market cooperative A; school
collection point B"},
 "identification": {"method": "wastewise-ml", "confidence": 0.93}, "contamination_pct": 1.5,
 "processed_by": "Demo micro-factory", "process": "wash, shred, float-sink, dry, sheet-press",
 "recycled_content": {"post_consumer_pct": 100, "pre_consumer_pct": 0, "basis": "mass",
"standard": "ISO 14021",
   "statement": "Contains 100 % post-consumer recycled HDPE by mass (self-declared, ISO
14021)."},
 "date": "2026-09-25"}
```


Key design choices (decided by Amish, 2026-09-25, RFE-DDR-001)

- **Plastics line as the reference micro-factory.** Plastics have the most open machine designs and the lowest entry cost. Aluminium remelting, rated high in the feasibility matrix, is a separate add-on bay with its own safety case, not part of the reference floor.
- **Two 40 ft container footprints.** A single 40 ft container is about 2.35 m wide inside, too narrow for safe aisles around a sheet press. Two side by side (or a shed of the same area) give 59.4 m².
- **PET sold as flake; HDPE and PP made into products.** This keeps the hot zone to the two polymers that small extruders and presses handle well.
- **Single-phase supply with staggered heating.** A heater interlock keeps the press heaters off while the shredder or the extruder runs. The STEM check of 2026-10-25 found that the morning loads still draw about 55 A from a 40 A supply when counted as currents; the options (three-phase, a longer day in three blocks, or less input) are with Amish (RFE-DEC-001 D1).
- **Passport schema v0.2.** Parent batch IDs, schema version, product form and a recycled content statement, issued as a new file beside v0.1.

Further choices accepted by Amish on 2026-09-25 (RFE-DDR-002):

- **Honest disposal count with an intake quality rule.** PR-008 counts process losses as well as sorting residue. A delivery whose sample shows more than 10 % non-recoverable residue is refused or charged a disposal fee (PR-016), because the site can refuse poor loads but cannot make losses disappear.
- **Derated 5 kW single-phase sheet press as the reference.** The published 15 kW, 400 V press is the option where three-phase exists. The derated press has almost no time margin (RFE-CAL-001 section 3).
- **1.5 kW fume fan.** It keeps margin as the filters load up (1.00 kW drawn with dirty filters).
- **Rented shed preferred.** The two-container shell is the fallback, with side walls removed only to a structural engineer's design.
- **License for the first release.** Documents under CC BY-SA 4.0, hardware under CERN-OHL-S-2.0 and scripts under MIT, applied when the first release is tagged. Until then the repo licenses are unchanged.

Two layout choices follow from these and are not separate decisions: one hot zone under two enclosing hoods, away from the intake and the product store; and passports issued at the desk, not at each machine.

Safety

> **Safety:** A micro-factory combines moving machinery, hot surfaces, fumes, water near electricity, sharp and contaminated input and a high fire load of plastic. These rules extend `docs/playbook/safety.md` and must be reviewed by a qualified safety professional before any site operates.

- **Shredder (moving machinery).** Fixed guards, a hopper interlock that stops the rotor when opened, an emergency stop within reach, lockout before clearing jams, and no hands or tools in the hopper. In the reverberant steel room a bare shredder would put the whole floor at about 95 dB(A), so the shredder sits in a lined acoustic enclosure with an interlocked feed chute and door (about 79 dB(A) over 8 h on paper, RFE-CAL-001 section 7). The shredder area stays a hearing protection zone until the level is measured. The reach from the 1.70 m chute mouth to the cutters was checked against ISO 13857 from the model: a straight collar fails (600 mm against 850 mm), so a cranked feed hood with a closed top sits on the collar and the feed goes in through a 460 x 120 mm slot facing the aisle, which gives 1,060 mm against the 850 mm that ISO 13857 Table 4 requires for openings up to 120 mm (RFE-CAL-001 section 7; slot lowered from 160 mm by the STEM check, RFE-DEC-001 D3); the enclosure door and the cage gates have self-closing hinges and open into the aisle only during locked-off maintenance or loading (decided by Amish, 2026-10-02).
- **Heat.** Extruder barrels and press platens run at about 190 to 200 °C (never above 230 °C for HDPE or 240 °C for PP). Insulated covers, heat-resistant gloves, face shields and a cool-down rule before maintenance. Never leave heaters on unattended.
- **Fumes.** Only identified HDPE and PP are melted. PVC is never heated (it releases hydrogen chloride), and polystyrene, PET and unknown plastics are not melted in the reference line. Extraction runs whenever a heater is on and for 30 min after, through two enclosing hoods sized for 0.5 m/s at their openings (RFE-CAL-001 section 6).
- **Electricity and water.** 30 mA RCDs on every circuit, splash-rated sockets near the wash tanks, a heater interlock that keeps the press heaters off while a motor runs, and an installation by a licensed electrician who confirms the supply against the input currents of the drives actually bought (RFE-CAL-001 section 5; RFE-DEC-001 D1).
- **Input hazards.** Needles, broken glass, cans and chemical containers arrive mixed in. Puncture-resistant gloves, a sharps container, and a rule that medical, chemical and hazardous waste is refused at the door.
- **Lithium cells.** Pulled out first at intake, never punctured or crushed, terminals taped and stored in a fire-safe container with dry sand. A cell in the shredder can start a fire.
- **Fire.** Plastic stock and flake are a high fire load. Three extinguishers (entrance, hot zone, product end), stock kept at least 1.0 m from the hot zone, clear exits, no smoking and a daily cleanup of dust and flake. Fine plastic dust can be combustible.
- **Wastewater and sludge.** Settled and disposed of through a licensed route, never discharged to drains or open ground.
- **People.** No workers under 18. PPE is supplied free. Manual handling limited to bags of 25 kg or less.
- **Structure.** A rented shed is preferred (RFE-DDR-002). Removing the adjoining side walls of two containers weakens them, so the container option is used only with reinforcement specified by a structural engineer.
- **Molten metal.** No aluminium is melted on the reference floor. The add-on bay needs its own safety case first (RFE-DDR-001 item 1; `docs/playbook/recipes/aluminium.md`).

Open questions

- [ ] Real target-plastic share and contamination of collected input in the first partner region (drives PR-007 and PR-008)
- [ ] Measured shredder, extruder and sheet press rates, and whether the decided 5 kW single-phase press can be built to hold a 55 min cycle (PR-006; RFE-DDR-002 item 9)
- [ ] Measured shredder sound power and enclosure insertion loss (PR-012)
- [ ] Air monitoring at the hot zone on a running site (PR-011)
- [ ] Local prices for products, flake, metals and paper, and local wages (economics model inputs)
- [ ] Passport fields that buyers actually need, and a form fast enough for PR-015
- [ ] Permits for a small plastics recycler in the first target region
- [ ] Whether collectors can supply loads at 10 % residue or less, the PR-016 threshold (PR-008)
- [x] First co-design partner and region (RFE-DDR-001 item 7): a waste-picker cooperative; the first candidate to approach is a waste-picker cooperative, for example SWaCH in Pune, India, reached through the WIEGO network; nothing is agreed (decided by Amish, 2026-10-02)